5.9 Differential Calculus on Banach Spaces

The main reference for this section is [Car71], along with [Lan12], in a generality which is almost certainly not required for the purposes of these notes.

5.9.1 Total Derivatives

Definition 5.9.1
*Let $U \subset X$ be an open neighbourhood of 0 and $\varphi : U \to Y$ a function. We say φ is **sublinear** if*

$$\forall \varepsilon > 0 \exists \delta \text{ s.t. } \|h\| < \delta \implies (h \in U) \wedge (\|\varphi(h)\| < \varepsilon \|h\|)$$

Note sublinear functions are closed under linear combinations, and the property is unchanged under equivalent norms.

Definition 5.9.2 (Total Derivative)
*Let $U \subset X$ be an open subset of a Banach space, Y a Banach space and $x \in U$. We say that a map $\varphi : U \to Y$ is differentiable at x if there exists a continuous linear map $D\varphi(x) \in L(X, Y)$ (**derivative** at x) such that*

$$\varphi(x + h) - \varphi(x) - D\varphi(x)(h) \text{ is sublinear in } h$$

*We say it is simply **differentiable** if this holds for every point $x \in U$ and write $D\varphi$ for the corresponding function $U \to L(X, Y)$.*

*If $D\varphi$ is also continuous then we say that φ is **continuously differentiable** or C^1.*

The following result shows that the total derivative is necessarily unique.

Proposition 5.9.3
Let $\alpha : X \to Y$ be a linear map which is also sublinear. Then $\alpha = 0$.

Proof. Consider $x \in X$ then for every $\varepsilon > 0$ there exists δ such that $\|h\| < \delta \implies \|\alpha(h)\| < \varepsilon \|h\|$. Define $\lambda := \frac{\delta}{2\|x\|}$. Then evidently $\|\alpha(\lambda x)\| < \varepsilon \|\lambda x\|$ whence $\|\alpha(x)\| < \varepsilon \|x\|$. As this holds for every ε we see $\|\alpha(x)\| = 0$ and $\alpha(x) = 0$. ∎

Proposition 5.9.4
Let $f : U \to Y$ be a differentiable map and $\alpha : Y \to Z$ a linear map. Then

$$D(\alpha \circ f)(x) = \alpha \circ Df(x)$$

Proposition 5.9.5
Let $U \subset K$ an open set and $\varphi : U \to Y$ a function. Then the following are equivalent

a) φ is (continuously) differentiable in the sense of (5.9.2)
b) φ is (continuously) differentiable in the sense of (5.5.1)

Furthermore we may identify the derivatives as follows

$$D\varphi(t)(\lambda) = \varphi'(t)\lambda$$

Proposition 5.9.6 (Chain Rule)
Let $U \subset X$ and $V \subset Y$ be open sets and $\varphi : U \to Y$ and $\psi : V \to Z$ (continuously) differentiable functions such that $\varphi(U) \subset V$. Then $\psi \circ \varphi$ is (continuously) differentiable and

$$(D(\psi \circ \varphi))(x) = (D\psi)(\varphi(x)) \circ (D\varphi)(x)$$

In the case $X = K$ we have

$$(\psi \circ \varphi)'(t) = (D\psi)(\varphi(t))(\varphi'(t))$$

Proposition 5.9.7 (Product Rule I)
Let $U \subset X$ and $V \subset Y$ open sets, $\varphi : U \to X'$, $\psi : V \to Y'$ (continuously) differentiable maps and $\Psi : X' \times Y' \to Z$ a C^1 bilinear map. Then define

$$(\varphi \times_\Psi \psi)(x, y) := \Psi(\varphi(x), \psi(y))$$

is (continuously) differentiable with derivative

$$D(\varphi \times_\Psi \psi)(x, y)(h, k) = \Psi(D\varphi(x)(h), \psi(y)) + \Psi(\varphi(x), D\psi(y)(k))$$

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Corollary 5.9.8
Let X, Y be normed vector spaces, $U \subset X$ an open subset and $\varphi, \psi : X \to Y$ (continuously) differentiable at x. Then both $\varphi + \psi$ and $\lambda\varphi$ are (continuously) differentiable at x and

$$D(\varphi + \psi)(x) = D(\varphi)(x) + D(\psi)(x)$$
$$D(\lambda\varphi)(x) = \lambda D(\varphi)(x)$$

If Y is also a Banach algebra then $\varphi \cdot \psi$ is (continuously) differentiable at x and

$$D(\varphi \cdot \psi)(x)(h) = D(\varphi)(x)(h) \cdot \psi(x) + \varphi(x) \cdot D(\psi)(x)(h)$$

Corollary 5.9.9 (Product Rule II)
Let $U \subset X$ be an open set, and $\varphi : U \to K$ and $\psi : U \to Y$ differentiable maps. Then

$$D(\varphi \cdot \psi)(x)(h) = D\varphi(x)(h)\psi(x) + \varphi(x)D\psi(x)(h)$$

If φ, ψ are C^1 then so is $\varphi \cdot \psi$.

Definition 5.9.10 (Higher-Order Derivatives)
*Let $f : U \to Y$ we say that f is n-times **continuously differentiable** (or C^n) if it is differentiable and Df is C^{n-1} (where C^0 means simply continuous). In this case for $k \leq n$ we may define the k-fold derivative inductively*

$$D^k f := D\big(D^{k-1} f\big) : U \longrightarrow L(X, \cdots, L(X, Y))$$

which is by definition continuous.

Proposition 5.9.11
Let $U \subset K$ be an open subset and $f : U \to Y$ a function. Then f is n-times differentiable in the sense of (...) if and only if it is n-times differentiable in the sense of (5.9.10). In this case

$$(D^n f)(x)(h_1, \cdots, h_n) = f^{(n)}(x) \cdot h_1 \cdots h_n$$

Proposition 5.9.12 (Mean Value Theorem)
Let $U \subset X$ be an open subset and $f : U \to Y$ a C^1 map. Suppose that $h \in B(x; r) \subset U$. Then

$$f(x + h) = f(x) + \int_0^1 Df(x + th)(h)\,\mathrm{d}t$$

Proof. Let $G(t) := f(x + th)$ be defined as a function on $[0, 1] \to Y$. Then by (5.9.6) $G'(t) = Df(x + th)(h)$ is continuous, because evidently $t \to x + th$ is continuously differentiable with scalar derivative h. The result follows by applying (5.8.20) to $G(t)$. ∎

5.9.2 Partial Derivatives

Lemma 5.9.13
Let $X_1, \cdots, X_n$ be Banach spaces and for every $x \in X$ define the injection

$$\lambda_j(x) : X_j \longrightarrow X_1 \times \cdots \times X_n$$
$$x' \to \big(x_1, \cdots, x_{j-1}, x', x_{j+1}, \cdots, x_n\big)$$

Then λ_j is differentiable at every $x_j \in X_j$ and $\big(D\lambda_j(x)\big)\big(x_j\big) = \mu_j$ where

$$\mu_j(h) = (0, \cdots, h, \cdots, 0)$$

Proof. We have $\lambda_j(x)(x' + h') - \lambda_j(x) = \mu_j(h')$ and μ_j is linear. ∎

Proposition 5.9.14 (Criteria for Total Differentiability)
Let $X_1, \cdots, X_n, Y$ be Banach spaces and $U \subset X := X_1 \times \cdots \times X_n$ an open subset. Consider a continuous map

$$\varphi : U \to Y$$

The following are equivalent

a) The map φ is continuously differentiable (with respect to the sup norm on the product)
b) For every $x \in U$ define $V_j := \lambda_j(x)^{-1}(U) \subset X_j$ an open neighbourhood of x_j
i) The map

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$$\varphi_j(x) := \varphi \circ \lambda_j(x) : V_j \to Y$$

is differentiable for all $j = 1 \cdots n$. Denote the j-th partial derivative at $x \in U$ by

$$D_j\varphi : U \to L\big(X_j, Y\big)$$
$$x_j \to h \to \big(D\varphi_j(x)\big)\big(x_j\big)(h)$$

ii) The map D_j is continuous for all $j = 1 \cdots n$

Furthermore in this case

$$D\varphi(x)(h) = \sum_{j=1}^{n} \big(D_j\varphi\big)(x)\big(h_j\big)$$

Proof. **a) $\implies$ b)** By the chain rule (5.9.6) and (5.9.13) we have φ_j is differentiable and

$$\big(D\varphi_j(x)\big)\big(x_j\big) = (D\varphi)(x) \circ \big(D\lambda_j(x)\big)\big(x_j\big) = (D\varphi)(x) \circ \mu_j$$

b) $\implies$ a) We consider the case $n = 2$. Observe

$$\begin{aligned}
\varphi(x_1 + h_1, x_2 + h_2) - \varphi(x_1, x_2) &= \varphi(x_1 + h_1, x_2 + h_2) - \varphi(x_1 + h_1, x_2) + \varphi(x_1 + h_1, x_2) - \varphi(x_1, x_2) \\
&= \int_0^1 D_2\varphi(x_1 + h_1, x_2 + th_2)(h_2)\,\mathrm{d}t + \int_0^1 D_1\varphi(x_1 + th_1, x_2)(h_1)\,\mathrm{d}t \\
&= D_2\varphi(x_1, x_2)(h_2) + D_1\varphi(x_1, x_2)(h_1) \\
&+ \int_0^1 [D_2\varphi(x_1 + h_1, x_2 + th_2)(h_2) - D_2\varphi(x_1, x_2)(h_2)]\,\mathrm{d}t \\
&+ \int_0^1 [D_1\varphi(x_1 + th_1, x_2)(h_1) - D_1\varphi(x_1, x_2)(h_1)]\,\mathrm{d}t
\end{aligned}$$

Then

$$\begin{aligned}
\left\| \int_0^1 [D_2\varphi(x_1 + h_1, x_2 + th_2)(h_2) - D_2\varphi(x_1, x_2)(h_2)]\,\mathrm{d}t \right\| &\leq \int_0^1 \|D_2\varphi(x_1 + h_1, x_2 + th_2)(h_2) - D_2\varphi(x_1, x_2)(h_2)\|\,\mathrm{d}t \\
&\leq \int_0^1 \|D_2\varphi(x_1 + h_1, x_2 + th_2) - D_2\varphi(x_1, x_2)\|\|h_2\|\,\mathrm{d}t \\
&\leq \int_0^1 \|D_2\varphi\|\|(h_1, th_2)\|\|h_2\|\,\mathrm{d}t \\
&\leq \|D_2\varphi\|\|h\|^2
\end{aligned}$$

and similarly for the second integral. Therefore φ is differentiable at (x_1, x_2) with derivative

$$D\varphi(x_1, x_2)(h) = D_1\varphi(x_1, x_2)(h_1) + D_2\varphi(x_1, x_2)(h_2)$$

as required.

We have $h = \sum_{j=1}^{n} \mu_j\big(h_j\big)$ and so by the first part

$$\begin{aligned}
D\varphi(x)(h) &= \sum_{j=1}^{n} \big(D\varphi(x)\big)\big(\mu_j\big(h_j\big)\big) \\
&= \sum_{j=1}^{n} \big(D\varphi_j(x)\big)\big(x_j\big)\big(h_j\big) \\
&= \sum_{j=1}^{n} \big(D_j\varphi\big)(x)\big(h_j\big)
\end{aligned}$$

■

Proposition 5.9.15
Let $U \subset X$ be an open set and

$$\varphi : U \to Y_1 \times \cdots \times Y_m$$

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Then φ is differentiable (resp. C^1) iff $\pi_i \circ \varphi$ is differentiable (resp. C^1) for $i = 1 \cdots m$.

Proposition 5.9.16 (Jacobian Matrix)
Let X, Y be finite-dimensional Banach spaces with bases $\{e_1, \cdots, e_n\}, \{f_1, \cdots, f_m\}$ such that $U \subset X$ is open and $\varphi : U \to Y$ a map. Then the following are equivalent

a) φ is C^1
b) For every $x \in U$, $j = 1 \cdots n$, $i = 1 \cdots m$ the map

$$g_{ij}(x) : V \to K$$
$$t \to (f_i^\vee \circ \varphi)(x + te_j)$$

is continuously differentiable on the open set $V = \{t \in K \mid x + te_j \in U\}$.

In this case define $\frac{\partial \varphi_i}{\partial e_j}(x) \coloneqq g_{ij}(x)'(0)$. Then the linear map $D\varphi(x)$ has the following matrix representation

$$J_\varphi(x) \coloneqq [D\varphi(x)] = \left(\frac{\partial \varphi_i}{\partial e_j}(x)\right)$$

*which we refer to as the **Jacobian Matrix**.*

When $Y = K$ then $\{1_K\}$ is a basis and we may simply write $\frac{\partial \varphi}{\partial e_j} : U \to K$ as the partial derivative.

Proof. Let $\psi_j : K \xrightarrow{\sim} \langle e_j \rangle$ be the canonical linear isomorphism and $\lambda_j(x) : \langle e_j \rangle \to X$ by the function defined in (5.9.13). Then

$$f_i^\vee \circ \varphi \circ \lambda_j(x) = g_{ij}(x) \circ \psi_j^{-1}$$

Suppose *b*) is satisfied. Because ψ_j^{-1} is continuously differentiable the hypotheses of (5.9.14) b) are satisfied and we conclude that $f_i^\vee \circ \varphi$ is C^1. Then by (5.9.15) so is φ.

Conversely if φ is C^1, then so is $f_i^\vee \circ \varphi$. And we may use the same identity and the chain rule to deduce $g_{ij}(x)$ is continuously differentiable.

By the same result we have

$$(f_i^\vee \circ D\varphi(x))(x_j) = D(f_i^\vee \circ \varphi)(x)(e_j) = D(f_i^\vee \circ \varphi \circ \lambda_j(x))(e_j)(e_j) = g_{ij}(x)'(0)e_j$$

which shows that $J_\varphi(x)$ has the required form (...). ∎

Proposition 5.9.17
Let X, Y, Z be finite-dimensional Banach spaces with bases $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$ respectively. Suppose $U \subset X, V \subset Y$ are open sets and $\varphi : U \to Y, \psi : V \to Z$ are C^1 maps such that $\varphi(U) \subset V$. Then $\psi \circ \varphi$ is C^1 and

$$J_{\psi \circ \varphi}(x) = J_\psi(\varphi(x)) J_\varphi(x)$$

5.9.3 Second Derivative

Definition 5.9.18
*For a map $\alpha : X \times X \to Y$ we say that $\alpha(v, w)$ is **sublinear** if for all $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$\|(v, w)\| < \delta \implies \|\alpha(v, w)\| < \varepsilon \|v\| \|w\|$$

where the norm on $X \times X$ is the product norm.

As before we have the following uniqueness property

Lemma 5.9.19
Let $\alpha : X \times X \to Y$ be a bilinear map which is also sublinear. Then $\alpha = 0$.

Proposition 5.9.20 (Second derivative is symmetric)
Let $f : U \to Y$ be a C^2 map. Then for all $x \in U$ we have $D^2 f(x)$ is the unique bilinear map $X \times X \to Y$ such that

$$f(x + v + w) - f(x + w) - f(x + v) + f(x) - D^2 f(x)(v, w) \text{ is sublinear}$$

Furthermore it is symmetric.

Proof. Uniqueness follows from (5.9.19) and from this it immediately follows that $D^2 f(x)$ is symmetric. Therefore we only need to show the required property. Let $v, w \in X$ be such that $x + tv + sw \in U$ for all $0 \leq s, t \leq 1$. Then define

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$$g(x) := f(x+v) - f(x)$$

By (...) applied to g and Df we find

$$\begin{aligned} g(x+w) - g(x) &= \int_0^1 [Df(x+v+tw)(w) - Df(x+tw)(w)]\,\mathrm{d}t \\ &= \int_0^1 \int_0^1 D^2 f(x+sv+tw)(v)\,\mathrm{d}s(w)\,\mathrm{d}t \\ &= D^2 f(x)(v,w) + \int_0^1 \int_0^1 [D^2 f(x+sv+tw)(v) - D^2 f(x)(v)]\,\mathrm{d}s(w)\,\mathrm{d}t \end{aligned}$$

Let $\psi(v,w)$ denote the integral then

$$\begin{aligned} \|\psi(v,w)\| &\le \int_0^1 \int_0^1 \left\|D^2 f(x+sv+tw)(v) - D^2 f(x)(v)\right\| \mathrm{d}s \|w\|\,\mathrm{d}t \\ &\le \int_0^1 \int_0^1 \left\|D^2 f(x+sv+tw) - D^2 f(x)\right\| \mathrm{d}s\,\mathrm{d}t \cdot \|v\|\|w\| \\ &\le \sup_{0\le s,t\le 1} \left\|D^2 f(x+sv+tw) - D^2 f(x)\right\| \|v\|\|w\| \end{aligned}$$

where the estimate is finite by continuity of $D^2 f$ and the Boundedness Theorem (5.7.8). Note that $\|sv+tw\| \le \|v\| + \|w\| \le 2\|(v,w)\|$. Therefore we may use the continuity of $D^2 f$ to show that ψ is sublinear as required. ∎

Proposition 5.9.21 (Symmetry of partial derivatives)
Let $X = X_1 \times \cdots \times X_n$, $\varphi : U \to Y$ be a C^2 map and $x \in U$. Then

$$D^2\varphi(x)\big((0,\cdots,h_i,\cdots,0),(0,\cdots,h_j,\cdots,0)\big) = D_i D_j \varphi(x)\big(h_i,h_j\big)$$

In particular

$$D_i D_j \varphi(x)\big(h_j,h_i\big) = D_j D_i \varphi(x)\big(h_i,h_j\big)$$

Proof. The first equation follows from (5.9.14) and the second from (5.9.20). ∎

Proposition 5.9.22 (Hessian Matrix)
Let X be a finite-dimensional Banach spaces with basis $\{e_1,\cdots,e_n\}$, $U \subset X$ an open subset and $\varphi : U \to K$ a C^2 map. Then the second order partial derivatives $\frac{\partial^2\varphi}{\partial e_i \partial e_j} := \frac{\partial}{\partial e_j}\frac{\partial\varphi}{\partial e_i} : U \to K$ exist for all $1 \le i,j \le n$.

Denote by

$$H_\varphi := \left(\frac{\partial^2\varphi}{\partial e_i \partial e_j}\right)_{1\le i,j\le n}$$

*the **Hessian Matrix** which is the matrix of the bilinear form $D^2\varphi$ with respect to the given basis, and in particular symmetric.*

5.9.4 Differentiation under Integral Sign

Proposition 5.9.23
Let $A \subset X$ be a compact subset of a normed vector space and $f : A \to Y$ a continuous map. Then f is uniformly continuous. That is for all $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\|x-y\| < \delta \implies \|f(x) - f(y)\| < \varepsilon \quad \forall x,y \in A$$

Proof. Given $\varepsilon > 0$ for each $x \in A$ by continuity there exists a real number $\delta(x) > 0$ such that $\|y-x\| < \delta(x) \implies \|f(y) - f(x)\| < \varepsilon$. Evidently the balls $B\big(x, \frac{\delta(x)}{2}\big)$ cover A, so we may find a finite subcover of balls centered at $x_1,\cdots,x_n$. Let $\delta := \min\Big(\frac{\delta(x_1)}{2},\cdots,\frac{\delta(x_n)}{2}\Big)$.

TODO ∎

Proposition 5.9.24
Let $I = [a,b]$ be a real interval, $U \subset X$ an open subset of a finite-dimensional normed vector space and $f : I \times U \to Y$ be a continuous map such that $D_2 f$ exists and is continuous. Define

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$$g(x) = \int_a^b f(t,x)\,\mathrm{d}t$$

Then g is differentiable on U with derivative given by

$$Dg(x)(h) = \int_a^b D_2 f(t,x)(h)\,\mathrm{d}t$$

Proof. Consider a point $x \in U$ with a compact neighbourhood $N \subset U$ (...). Consider $\lambda \in L(X,Y)$ determined by

$$\lambda(h) = \int_a^b (D_2 f)(t,x)(h)\,\mathrm{d}t$$

Then by (5.9.12)

$$\begin{aligned} g(x+h) - g(x) - \lambda(h) &= \int_a^b [f(t,x+h) - f(t,x) - (D_2 f)(t,x)(h)]\,\mathrm{d}t \\ &= \int_a^b \left[\int_0^1 (D_2 f)(t,x+uh)(h)\,\mathrm{d}u - (D_2 f)(t,x)(h)\right]\mathrm{d}t \\ &= \int_a^b \left[\int_0^1 [(D_2 f)(t,x+uh)(h) - (D_2 f)(t,x)(h)]\,\mathrm{d}u\right]\mathrm{d}t \end{aligned}$$

For some δ_1 we have $B(x,\delta_1) \subset N$. Therefore if $\|h\| < \delta_1$ then $x + uh \in N$ for all $u \in [0,1]$. Observe that

$$\|(t,x+uh) - (t,x)\| = \|(0,uh)\| = |u|\|h\|$$

By uniform continuity (5.9.23) on the compact set $[a,b] \times N$ (4.1.115) there is some δ_2 such that

$$(x+uh) \in N \wedge |u|\|h\| < \delta_2 \implies \|(D_2 f)(t,x+uh) - (D_2 f)(t,x)\| < \varepsilon$$

Therefore for $\|h\| < \min(\delta_1, \delta_2)$ we have

$$\begin{aligned} \|g(x+h) - g(x) - \lambda(h)\| &\leq \int_a^b \int_0^1 \|(D_2 f)(t,x+uh)(h) - (D_2 f)(t,x)(h)\|\,\mathrm{d}u\,\mathrm{d}t \\ &\leq (b-a)\|h\|\varepsilon \end{aligned}$$

By definition this means $g(x)$ is differentiable with the given derivative. ∎

Corollary 5.9.25
Let $I = [a,b]$ be a real interval, $U \subset K$ an open subset of a complete valued field K and $f(t,x) : I \times U \to Y$ a continuous map such that $\frac{\partial f}{\partial x} : I \times U \to K$ exists and is continuous. Define

$$g(x) = \int_a^b f(t,x)\,\mathrm{d}t$$

Then g is differentiable on U (in the sense of ....) and

$$g'(x) = \int_a^b \frac{\partial f}{\partial x}(t,x)\,\mathrm{d}t$$

Proof. This follows because $\frac{\partial f}{\partial x} = (D_2 f)(x)(1_K)$ and $g'(x) = (Dg)(x)(1_K)$. ∎

5.9.5 Taylor's Theorem

Lemma 5.9.26 (Integration by Parts)
Let $U : [0,1] \to K$ and $V : [0,1] \to X$ be C^1 functions. Then

$$\int_0^1 U'(t)V(t)\,\mathrm{d}t = U(1)V(1) - U(0)V(0) - \int_0^1 U(t)V'(t)\,\mathrm{d}t$$

Proof. Let $H(t) := U(t)V(t)$ then by (5.9.9) $H'(t) = U'(t)V(t) + U(t)V'(t)$ is continuous. The result then follows from (5.8.20). ∎

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Corollary 5.9.27 (Integration by Parts II)
Let $G : [0,1] \to X$ be a C^1 function. Then

$$\int_0^1 G(t)\,\mathrm{d}t = G(0) + \int_0^1 (1-t)G'(t)\,\mathrm{d}t$$

Proposition 5.9.28 (Taylor's Theorem)
Let $U \subset X$ be an open set and $f : U \to Y$ a C^2 map. Suppose $x \in U$, $B(x;r) \subset U$ and $\|h\| < r$. Then

$$f(x+h) = f(x) + Df(x)(h) + \int_0^1 (1-t)D^2 f(x+th)(h,h)\,\mathrm{d}t$$

Proof. Let $G(t) = Df(x+th)(h) : [0,1] \to Y$ then

$$G(t) = \mathrm{ev}_h \circ Df \circ (x+th)$$

By (5.9.4)

$$D(\mathrm{ev}_h \circ Df)(x) = \mathrm{ev}_h \circ D^2 f(x)$$

whence by (5.9.6) $G(t)$ is C^1 with scalar derivative

$$G'(t) = (\mathrm{ev}_h \circ D^2 f(x+th))(h) = D^2 f(x+th)(h,h)$$

Recall by the Mean Value Theorem (5.9.12) and Integration by Parts (5.9.27)

$$\begin{aligned} f(x+h) &= f(x) + \int_0^1 G(t)\,\mathrm{d}t \\ &= f(x) + G(0) + \int_0^1 (1-t)G'(t)\,\mathrm{d}t \end{aligned}$$

which is the required result. ∎

Corollary 5.9.29 (Scalar form of Taylor's Theorem)
Let X be a finite-dimensional Banach space with basis $\{e_1, \cdots, e_n\}$, $U \subset X$ an open subset and $f : U \to K$ a C^2 map. Suppose $x \in U$ and $y \in B(x;r) \subset U$ then

$$f(y) = f(x) + \sum_{i=1}^n (y_i - x_i)\frac{\partial f}{\partial e_i}(x) + \sum_{i,j=1}^n (y_i - x_i)(y_j - x_j)R_{ij}$$

with

$$R_{ij}(y;x) = \int_0^1 (1-t)\frac{\partial^2 f}{\partial e_i \partial e_j}(x + t(y-x))\,\mathrm{d}t$$

If f is C^{2+p} (resp. C^∞) then R_{ij} is C^p (resp. C^∞) on $B(x;r)$.

5.9.6 Differential Forms on Banach Spaces

Definition 5.9.30
Let X, Y be Banach spaces and $U \subset X$ an open subset. A continuous mapping

$$\omega : U \longrightarrow L_a^p(X,Y)$$

*is called a **differential form of degree** p or a **differential p-form***

We say that ω is n-times continuously differentiable if it is so with respect to the canonical norm on $L_a^p(X,Y)$.

Proposition 5.9.31
Let X be a finite-dimensional vector space with basis $e_1, \cdots, e_k$ and $U \subset X$ an open subset. For each $i = 1 \cdots k$ define the mapping

$$\mathrm{d}x_i : U \to L_a^1(X,K)$$

which assigns the dual map $e_i^\vee$. Then for every differential p-form ω we have

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$$\omega(x) = \sum_{i=1}^{n} c_i(x)\,\mathrm{d}x_i$$

for some unique functions $c_i : U \to K$.

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